

Trainings ▾

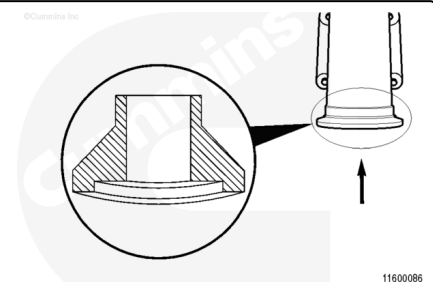
General Information

The manifold parts and mounting hardware are **not** the same for all K38 and K50 engines. Because the parts are different, procedures are described for installation of the exhaust manifolds for:

- KT38 and KTA38 Dry Exhaust Manifolds
- KTTA38 Dry Exhaust Manifolds
- KT38 and KTA38 Dry Exhaust Manifolds with Center Mount Turbochargers (Outboard aftercoolers)
- KT38, KTA38, and KTA50 Wet Exhaust Manifolds
- KTA50 Dry Exhaust Manifolds
- KTTA50 Dry Exhaust, Log Type Manifolds
- KTA50, KTTA50, K1500E, K1800E and K2000E Dry Exhaust, Pulse Type Manifolds
- K38 and K50 Bellows Manifolds.

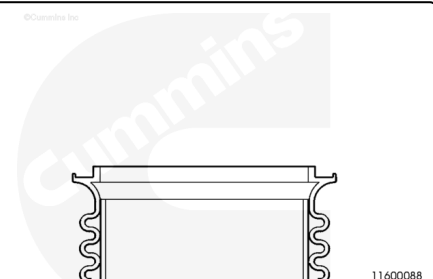
The marine exhaust system is used on the K38 M0/M1/M2/D(M1) and the K50M2/D(M1) engines.

The marine exhaust manifold side profile is shown.



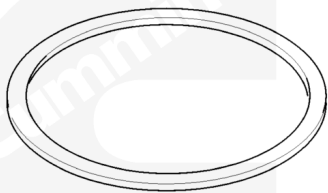
The marine exhaust manifold bellows has a different design than the industrial exhaust manifold bellows.

The marine exhaust manifold bellows is shown.



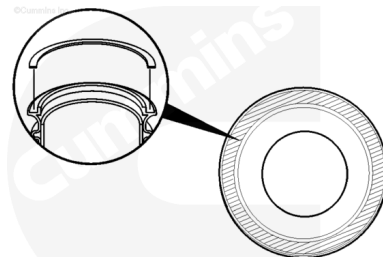
The exhaust manifold bellows gasket, Part Number 3966350, is located in the end of the bellows.

The bellows will **not** seal without the gasket in place.



11600089

The illustration shows the marine exhaust manifold bellows with the gasket fitted in place.



11600090

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before servicing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

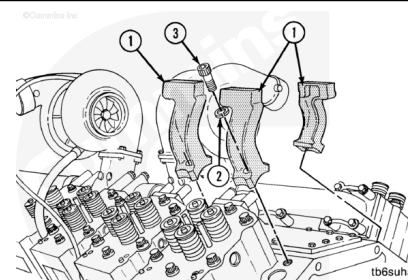
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

- Remove the turbochargers. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html>)
- Remove the air inlet and crossover connections. Refer to Procedure 010-020 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-020-tr.html>)
- Remove the wet manifold coolant supply tubes. Refer to Procedure 011-016 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html>)
- Remove the center mounted aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html>)
- Remove the turbochargers. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr-gc.html>)
- Remove the air inlet and crossover connections (Center-Mount Aftercooler). Refer to Procedure 010-020 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-020-tr.html>)
- Remove the wet manifold coolant supply tubes. Refer to Procedure 011-016 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-016-tr-gc.html>)
- Remove the center mounted aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr-gc.html>)

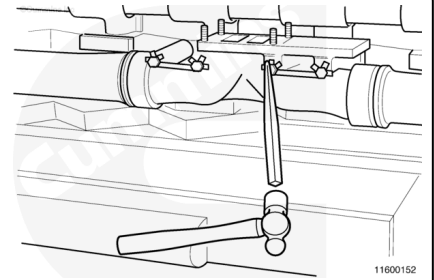
On KTTA engines, remove the low pressure turbocharger support brackets.



Remove

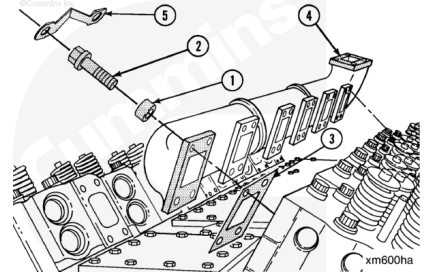
Use a drift and hammer to bend the lock plates away from the capscrew heads.

Note : All engines do not contain this type of lockplate. Some engines contain a lockplate that snaps over the capscrew head. Some engines have locking capscrews that do not require lockplates.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The dry manifold shown is a typical type. The exhaust manifolds for the K38 and K50 engines are different. The disassembly procedure, however, is the same.

There are four capscrews at each cylinder head exhaust port.

Remove the lockplates (5).

Remove the capscrews and spacers.

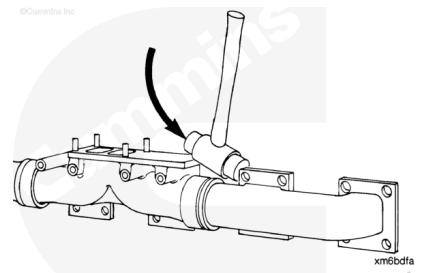
Remove both exhaust manifolds.

Remove the gaskets (3).

Discard the gaskets.

Disassemble

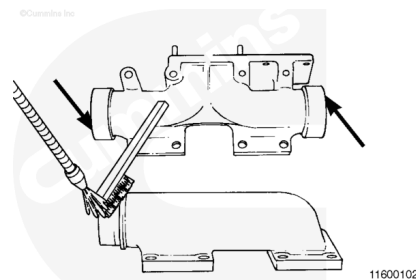
Use a mallet to separate the manifold sections.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



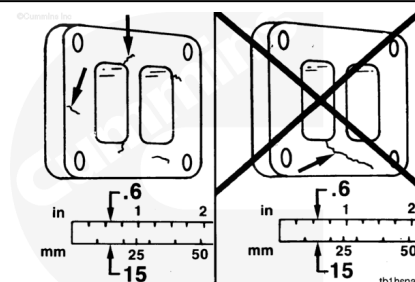
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Use steam or solvent to clean the manifold.

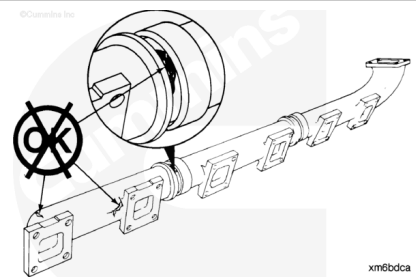
Use a wire brush to remove any scale from the inside and outside diameters of the sealing joints.

Use 240 grit emery cloth and clean all of the exhaust manifold gasket surfaces.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Note : This procedure applies to all K38 and K50 dry manifolds. **Only one style is illustrated.**

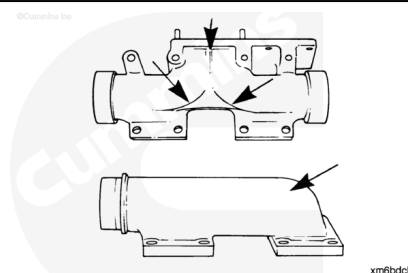
Check the manifold for cracks, warping, or leaks at the joints.

If any of the sections leak or are damaged, the manifold section **must** be replaced.

Use solvent or steam. Clean the parts.

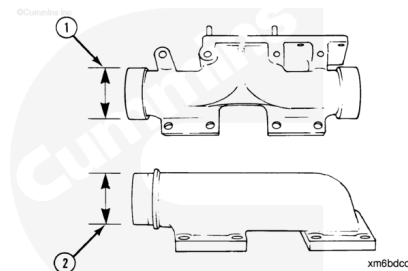
Check for cracks in the areas shown.

Check all turbocharger mounting studs for damage.



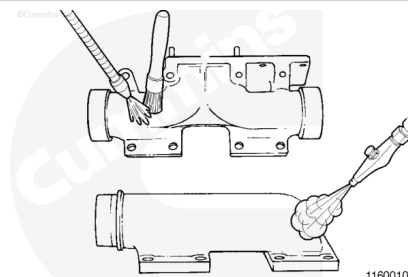
If a manifold section is to be installed again, measure the inside diameter (1) and the outside diameter (2) of the connecting manifold section.

The inside diameter can be a maximum of 0.063 mm [0.0025 in] larger than the outside diameter. The inside diameter and the outside diameter can be the same size.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



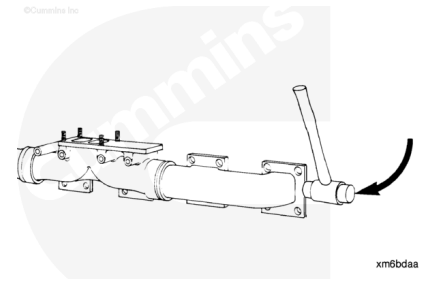
Note : This procedure applies to all K38 and K50 dry manifolds. **Only one style is illustrated.**

Use solvent or steam to clean the parts.

Assemble

Use an anti-seize compound to lubricate the outside diameter of the sections in the area that slides into the connecting manifold section.

Use a mallet to install the section to the connecting section.



Install

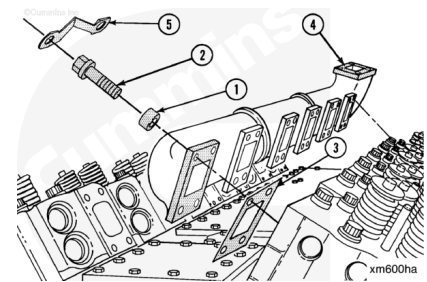
KT38 and KTA38 with Dry Exhaust Manifolds

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesive can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.



Note : The manifolds can be installed on either engine bank.

Note : Install the mounting capscrews (2) that are 76 mm [3 in] in length and spacers (1) that are 42.4 mm [1.67 in] in length for all applications except generator sets. Install capscrews that are 38 mm [1.5 in] in length and hardened washers for manifolds on generator set applications.

Install the spacers (1) on the capscrews (2). Apply an anti-seize compound on the threads of the capscrews.

The gasket **must** be installed so that the raised bead on the gasket is toward the manifold.

Install the gaskets (3), manifold (4), and capscrews. Check to be sure the gasket is aligned properly with the manifold and the cylinder head.

Torque Value:

38 mm [1.5 in] capscrew

1. 60 n•m [44 ft-lb]

Torque Value:

76 mm [3 in] capscrew with N stamped on head

1. 55 n•m [41 ft-lb]

Torque Value:

76 mm [3 in] capscrew without N stamped on head

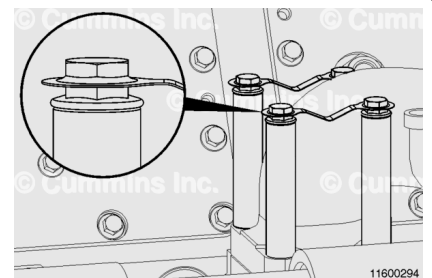
1. 80 n•m [59 ft-lb]

Note : Capscrews without the N stamped on the head have special threads to lock the capscrew and are **not** available for service. The torque value is included in case they are used again.

Note : Lockplates are **not** required on capscrews with locking threads.

⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.



New Part Number 3628969 **must** be used in place of Part Number 3043912.

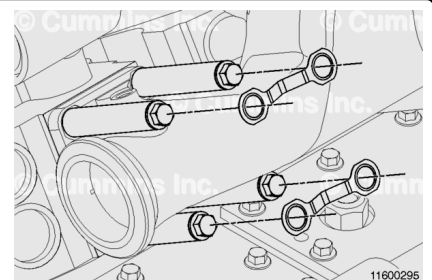
Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

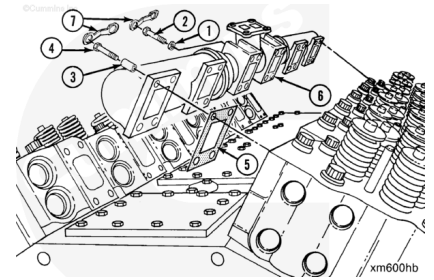
Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

Use service tool, Part Number 3163500, for correct installation height.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesive can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.

Note : The manifolds can be installed on either engine bank.

Install the mounting capscrews (2) that are 114 mm [4.5 in] in length and hardened washers (1) for the center section of the manifold. Install mounting capscrews (4) that are 76 mm [3 in] in length and the spacers (3) that are 42.4 mm [1.67 in] for the end sections.

Install the washers (1) on the capscrews (2). Install the spacers (3) on the capscrews (4). Apply an anti-seize compound on the threads of the capscrews.

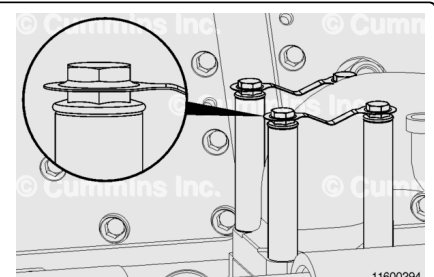
The gasket **must** be installed so that the raised bead on the gasket is toward the manifold.

Install the gaskets (5), the manifold (6), and the capscrews. Check to be sure the gasket is aligned properly with the manifold and the cylinder head.

Torque Value: 55 n·m [41 ft·lb]

⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.



New Part Number 3628969 **must** be used in place of Part Number 3043912.

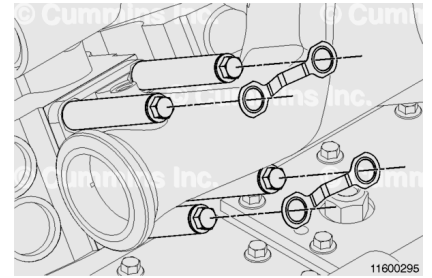
Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

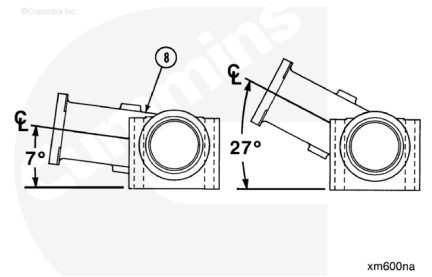
Use service tool, Part Number 3163500, for correct installation height.



KT38 and KTA38 with Dry Exhaust Manifolds

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesive can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.

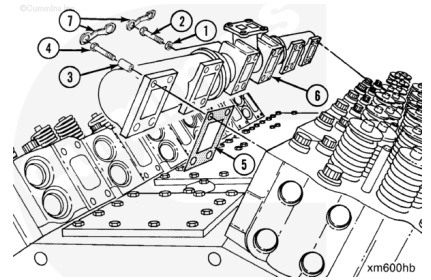
Note : The following information applies to center-mount turbochargers with outboard aftercoolers **only**.

The manifold center section **must** be installed on the correct engine bank for the engine exhaust piping to fit the application.

If the exhaust is to be toward the rear of the engine, the center section (8) with the smallest angle **must** be installed on the right bank.

If the exhaust is to be toward the front of the engine, the center section (8) with the smallest angle **must** be installed on the left bank.

Install the mounting capscrews (2) that are 114 mm [4.5 in] in length and hardened washers (1) for the center section of the manifold. Install the mounting capscrews (4) that are 76 mm [3 in] in length and the spacers (3) that are 42.4 mm [1.67 in] for the end sections.



Install the washers (1) on the capscrews (2). Install the spacers (3) on the capscrews (4). Apply an anti-seize compound on the threads of the capscrews.

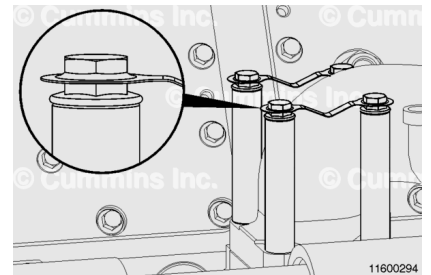
Note : The gasket must be installed so the raised bead on the gasket is toward the manifold.

Install the gaskets (5), manifold (6), and capscrews. Check to be sure the gasket is aligned properly with the manifold and the cylinder head.

Torque Value: 55 n•m [41 ft-lb]

⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.



New Part Number 3628969 **must** be used in place of Part Number 3043912.

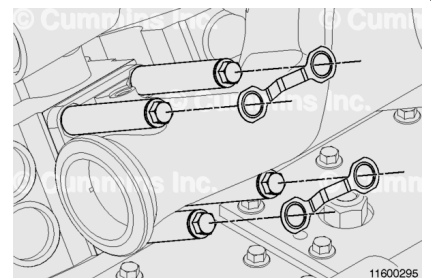
Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

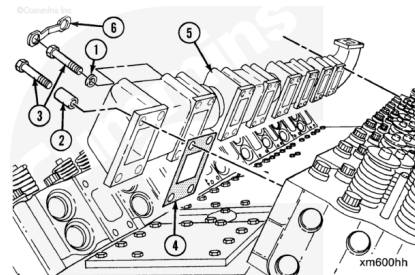
Use service tool, Part Number 3163500, for correct installation height.



KTA50 Dry Exhaust Manifolds

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesive can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.

Note : The manifolds can be installed on either engine bank.

Install capscrews (3) that are 140 mm [5.5 in] in length. Install hardened washers (1) on all the capscrews except at the end of the closed-end section. Install spacers (2) that are 105.64 mm [4.159 in] in length on the capscrews at the end of the closed-end section.

Note : Older engines contained a 38 mm [1.5 in] capscrew and a hardened washer (1) at the end of the closed-end section. These capscrews had special threads and are no longer available for service. If the older capscrew is replaced, the capscrew (3) and spacer (2) must be installed.

Install the hardened washers (1) and spacers (2) on the capscrews (3). Apply an anti-seize compound on the threads of the capscrews.

Note : The gasket must be installed so the raised bead on the gasket is toward the manifold.

Install the gaskets (4), manifold (5), and capscrews. Check to be sure the gasket is aligned properly with the manifold and the cylinder head.

Torque Value:

140 mm [5.5 in] capscrew with N stamped on head
1. 55 n•m [41 ft-lb]

Torque Value:

140 mm [5.5 in] and 38 mm [1.5 in] capscrews without N stamped on head

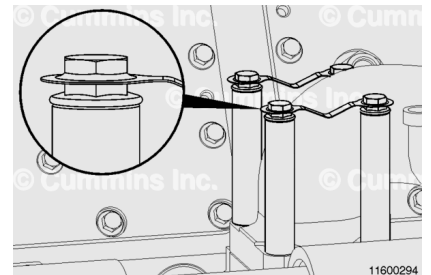
1. 80 n•m [59 ft-lb]

Note : Capscrews without the N stamped on the head have special threads to lock the capscrew and are no longer available for service. The torque value is included in case they are used again.

Note : Lockplates are not required on capscrews with special threads.

⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.



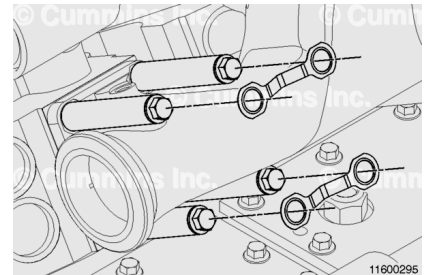
Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

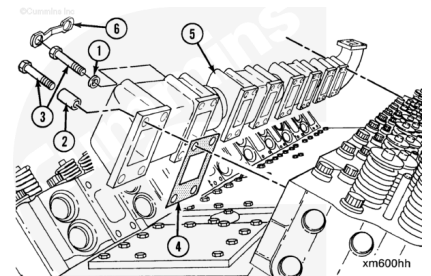
Use service tool, Part Number 3163500, for correct installation height.



KTТА50 Dry Exhaust, Log Type Manifolds

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesive can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.

Note : The manifolds can be installed on either engine bank.

Install capscrews (3) that are 140 mm [5.5 in] in length. Install hardened washers (1) on all the capscrews except at the end of the closed-end section. Install spacers (2) that are 105.64 mm [4.159 in] in length on the capscrews at the end of the closed-end section.

Older engines contained a 38 mm [1.5 in] capscrew and a hardened washer (1) at the end of the closed-end section. These capscrews had special threads and are no longer available for service. If the older capscrew is replaced, the capscrew (3) and spacer (2) **must** be installed.

Install the hardened washers (1) and spacers (2) on the capscrews (3). Apply an anti-seize compound on the threads of the capscrews.

Note : The gasket **must** be installed so the raised bead on the gasket is toward the manifold.

Install the gaskets (4), manifold (5), and capscrews. Check to be sure the gasket is aligned properly with the manifold and the cylinder head.

Torque Value:

140 mm [5.5 in] capscrew with N stamped on head
1.55 n•m [41 ft-lb]

Torque Value:

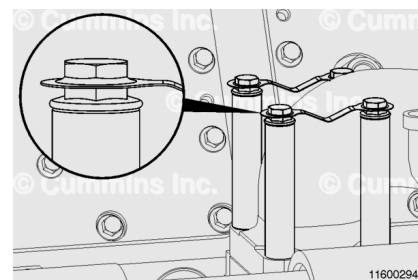
140 mm [5.5 in] and 38 mm [1.5 in] capscrews without N stamped on head
1.80 n•m [59 ft-lb]

Note : Capscrews without the N stamped on the head have special threads to lock the capscrew and are no longer available for service. The torque value is included in case they are used again.

Note : Lockplates are not required on capscrews with special threads.

⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.



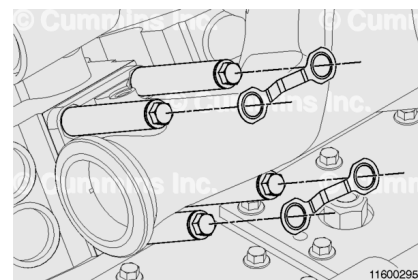
Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

Use service tool, Part Number 3163500, for correct installation height.

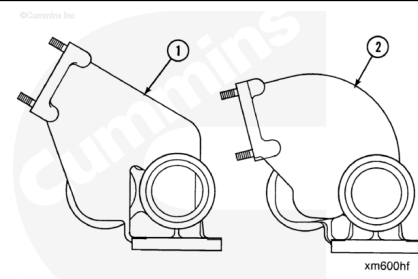


K50 Dry Exhaust, Pulse Type Manifolds

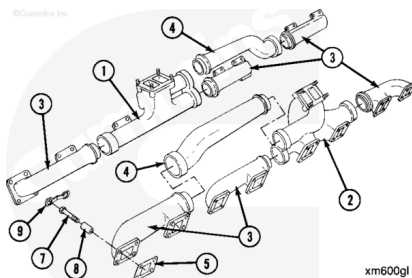
This procedure applies to the KTA50, KTTA50, K1500E, K1800E, 2000E manifolds also.

KTA50 engines with pulse-type manifolds contain a straight center section (1) on one bank and a curved center section (2) on the other bank. The manifold center sections **must** be installed on the correct engine bank for the engine exhaust piping to fit the application. If the exhaust is to be toward the rear of the engine, the manifold with the straight center section (1) **must** be installed on the left bank. If the exhaust is to be toward the front of the engine, the straight center section **must** be installed on the right bank.

KTTA50 engines with pulse-type manifolds contain a straight center section (1) in both banks. These manifolds can be installed on either engine bank.

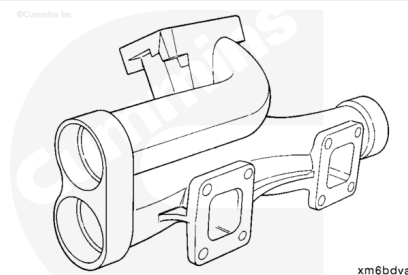


A variety of changes were made to the K50 pulse-type exhaust manifolds (1, 2, and 3) and manifold mounting parts (7, 8, and 9).

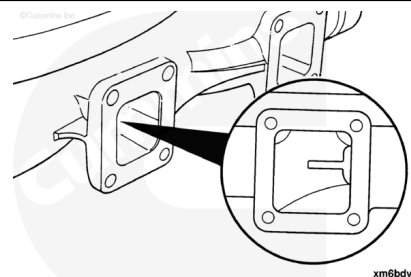


A revised straight center section, Part Number 3630241, replaced the previous straight center section, Part Number 3626757 or 3628682.

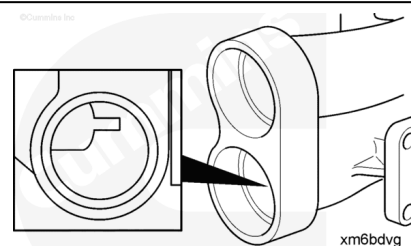
This revised straight center section is to be used on non-uprate K50 engines that require a pulse manifold (example: KTA50C-1600 engine).



This manifold, Part Number 3630241, can be identified by a notch in the exhaust port nearest to the two exhaust inlets or double end. The previous manifold, Part Number 3626757 or 3628682, did **not** have a notch.

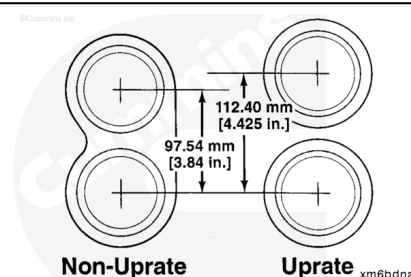


The notch can also be seen in the lower exhaust inlet.

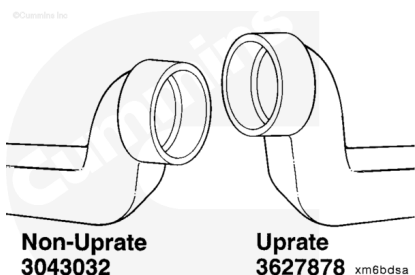


Uprate construction K50 exhaust manifold center sections require a higher thermal fatigue strength material (Ni-Resist).

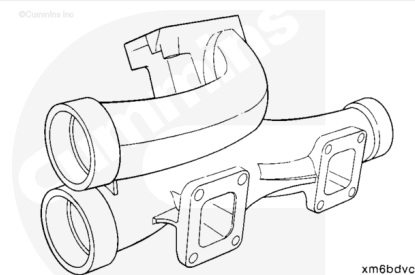
The distance between the two exhaust inlets or the double end has increased.



A dogleg exhaust manifold section, Part Number 3627878, was released for use on uprate K50 construction engines to accommodate this increased distance.

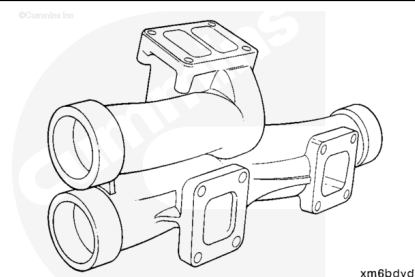


The uprate construction straight center section, Part Number 3629532, is to be used on the K2000E, KTTA50-C1800-B version, and on the left bank of the K1800E.



The construction uprate curved center section is to be used on the right bank of the K1800E.

The K1800E right-bank center section allows for rear-out exhaust.

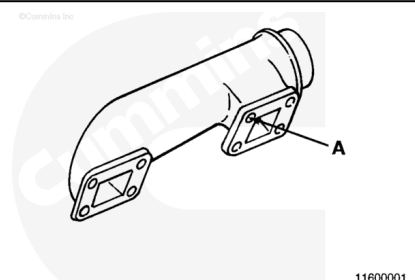


Engine Serial Number First for Sections	
Serial Number	Section Part Number
33122641	3627877
33122641	3627878
33124817	3630241
*	3629532

* Engine serial number first build date is March 1994.

The manifolds have been changed by increasing the size of the dowel hole (A) to the same diameter, 15 mm [0.591 in], as the other seven mounting holes within each section.

The smaller dowel hole was used to locate the manifold front to rear to make sure the slip joint was properly positioned.

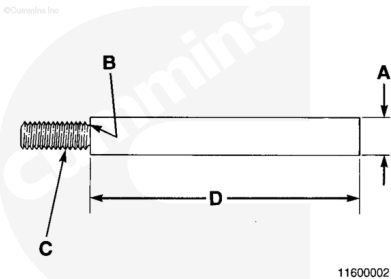


⚠ CAUTION ⚠

Guide pins or studs used on the manifolds will not be positioned properly, causing the manifold cylinder head capscrews to break.

Use locating pins to position the new manifolds properly. Use one pin in each section.

The diameter of the pin provides the proper manifold position and alignment.



The locating pin dimensions are listed below so they can be locally manufactured.

(A) Diameter:

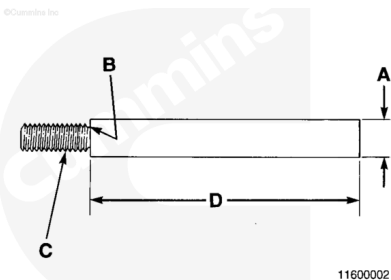
13.8 mm to 14.0 mm [0.543 in to 0.551 in].

(B) Undercut to thread root diameter

(C) Thread size 0.438 in x 14 in x 0.50 in long

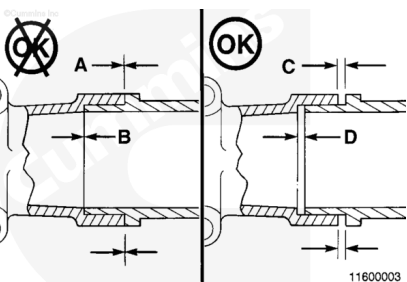
(D) Unthreaded length 99.2 mm [3.91 in].

Note : Wrench flats are optional on unthreaded end.



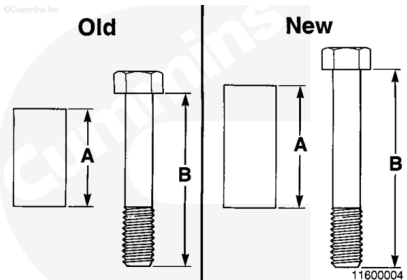
Inadequate clearance between exhaust manifold sections (A and B) will result in failed capscrews, exhaust manifolds, or cylinder heads. Use guide pins or studs to make sure the manifolds have proper clearance (C and D).

The typical manifold clearance (C) ranges from 4.0 to 9.5 mm [0.158 to 0.374 in].



Note : The longer capscrews and spacers can be used in all positions except the left-bank front without any component changes. Installation of longer capscrews and spacers in the left-bank front position requires a new thermostat housing support and water transfer tube. New capscrews and spacers can be used on any K50 pulse-type manifold part number.

Longer exhaust manifold mounting capscrews and spacers provide improved gasket sealing while permitting slippage or movement at the cylinder head and exhaust manifold interface.

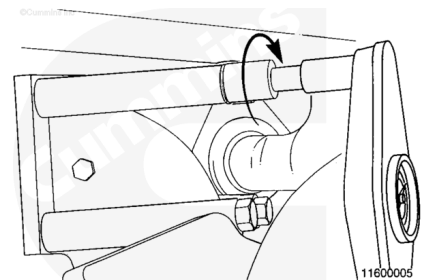


Part Name	Old Dimension	New Dimension
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Spacer-A	42.4 mm [1.67 in]	105.6 mm [4.16 in]
Spacer-B	76.2 mm [3.00 in]	139.7 mm [5.50 in]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use gasket adhesives. The adhesives can cause the molybdenum disulfide coating on the gasket to lose its effectiveness and cause a gasket or capscrew failure.

Apply an anti-seize compound on the threads of the capscrews.

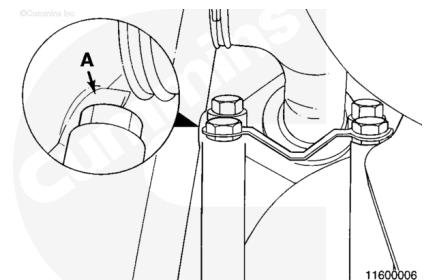
Note : The gasket must be installed so the raised bead on the gasket is toward the manifold.

Install the gaskets, manifold, and mounting capscrews and spacers. Check to be sure that the gasket is aligned properly with the manifold and the cylinder head.

Torque Value: 55 n•m [41 ft-lb]

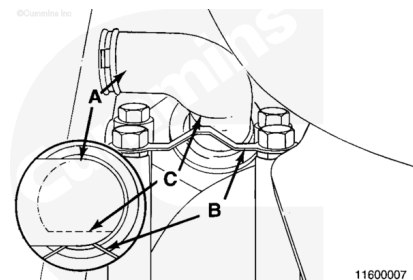
Note : When installing the lockplate over the capscrew head, install the plate to approximately one-half of the head height. Installing the lockplate until it bottoms will cause it to break.

In order to obtain clearance (A) for the longer capscrews and spacers, three other component changes have been made.



- 1. A clearance (A) has been cast into the thermostat housing support to permit wrench and lockplate clearance.
- 2. The water transfer tube (A) from the front of the V-cavity to the thermostat housing support interferes with the capscrew lockplate (B).

A flat has been added to the water transfer tube (A) to help prevent interference with the capscrew lockplate (B) when the longer spacers are used. The dotted line (C) represents the flat added to the tube.

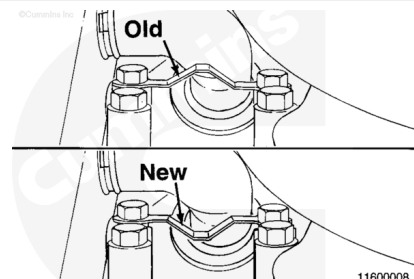


Note : Do not install the old lockplate, Part Number 3043912, upside down to clear the water tube. The lockplate will either break or not be retained properly on the capscrew head.

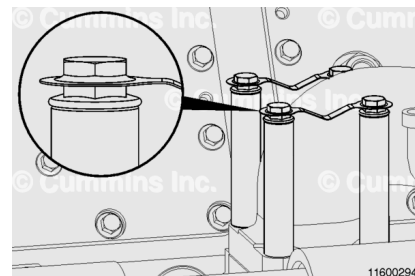
- 3. To increase the water-tube-to-lockplate clearance even more, a new lockplate was released. The new lockplate has the V-bend pointed downward toward the manifold, rather than upward.

Use service tool Part Number 3163500 to make sure the lockplates are installed correctly.

These changes were introduced in November 1994 (ESN first 33128992).



Use service tool, Part Number 3163500, to make sure the lockplates are installed correctly.

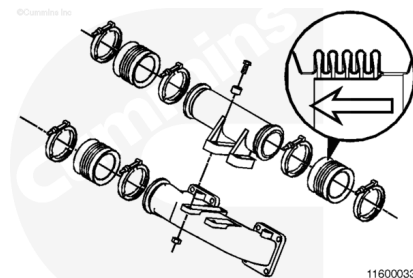


K38 and K50 Bellows Manifold

Install the bellows to both sides of the bolt-together manifold sections. Securing the bellows with V-band clamps, install the clamps with the tightening screw on the topside of the exhaust facing inboard.

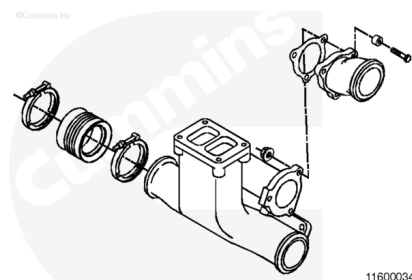
Note : Bellows must be installed correctly to prevent damage from the gas flow.

The arrow in the picture indicates the correct direction of gas flow through the bellows.

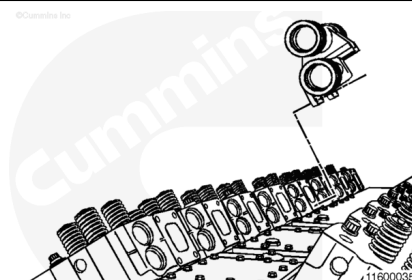


Install additional bellows to the single inlet side of the center turbocharger mounting section, securing the bellows V-clamp, as above.

Install the exhaust outlet adapter with the gasket.

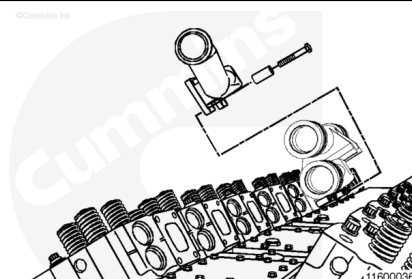


Starting on the left bank of the engine, install gaskets of the correct thickness and install the bolt-together pulse section to cylinder heads 5 and 6. Leave the clamps loose at this point.

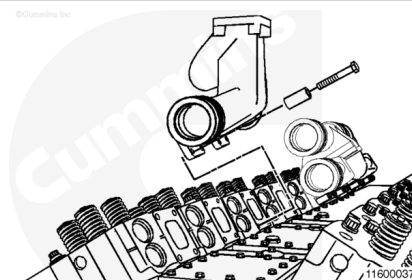


After installing the correct thickness gaskets, install the rear exhaust section while securing it to the bellows of the bolt-together section just installed.

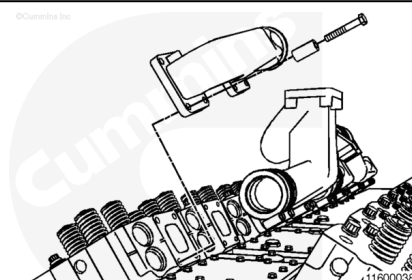
If the rocker housings are on the engine, then it will be necessary to loosen the bolt-together pulse section capscrews to allow the V-clamp to be fitted. Leave the capscrews loose at this point.



Install the center turbocharger mounting section while securing it to the two bellows of the bolt-together pulse section. Leave the main securing capscrews loose at this point.



Install the final end section of the exhaust system, leaving the capscrews loose at this point.



Remove two upper capscrews from the center sections of the exhaust system. Make sure the exhaust section is centered with the tapped holes in the cylinder heads.

Tighten the main securing capscrews.

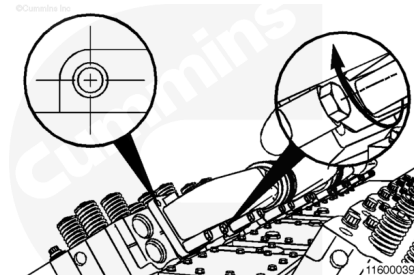
Torque Value: 54 n•m [40 ft-lb]

Move either direction to the next section and remove two capscrews. Center the exhaust manifold over the tapped holes.

Tighten the main capscrews.

Torque Value: 54 n•m [40 ft-lb]

Note : It is important that each section of the exhaust system be centralized to the tapped holes in the cylinder head before tightening mounting capscrews and V-band clamps. It is also important that the bellows be installed correctly for the gas flow direction.



Position the V-band clamps at approximately 110 degrees to the exhaust mounting face.

Torque Value:

Tighten the marine application V-band clamp

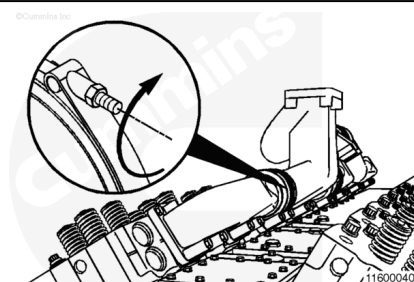
1. 17 n•m [150 in-lb]

Torque Value:

Tighten the industrial application V-band clamp

1. 5 n•m [44 in-lb]

Repeat the above installation instructions on right bank.

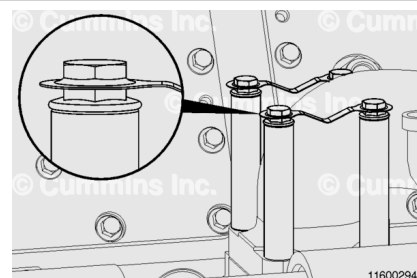


⚠ CAUTION ⚠

Do not install the lockplate with the incorrect orientation. The lockplate will either break or not be retained properly on the capscrew head.

Note : When installing the lockplate over the capscrew head, install the plate approximately one-half of the capscrew head height. Installing the lockplate until it bottoms will cause it to deform.

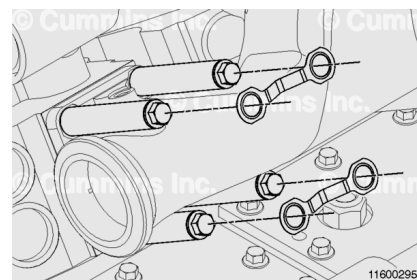
The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.



Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank of exhaust manifolds.

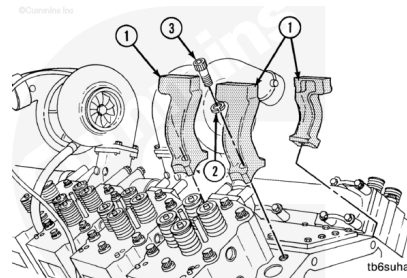
Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplates are fitted correctly. For all engines after ESN 33128992, the v-bend is facing toward the manifold.

Use service tool, Part Number 3163500, for correct installation height.



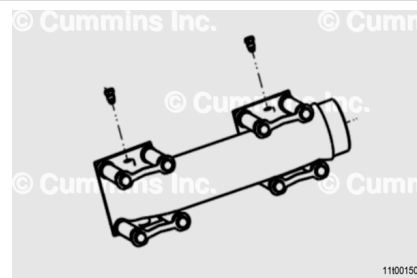
Finishing Steps

On KTTA engines, install the low pressure turbocharger support brackets.



On QSK engines without EGTS sensors, fit threaded plugs to the EGTS ports on exhaust manifolds.

Torque Value: 34 n•m [25 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the center mounted aftercooler. Refer to Procedure 010-002 in Section 10.

(/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)

- Install the turbochargers. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Install the wet manifold coolant tubes. Refer to Procedure 011-016 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Operate the engine and check for leaks.
- Install the center mounted aftercooler. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr-gc.html)
- Install the turbochargers. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr-gc.html)
- Install the wet manifold coolant tubes. Refer to Procedure 011-016 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-016-tr-gc.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html)
- Operate the engine and check for leaks.